

Technical PRD: Healthcare Innovation Scouting Platform

1. System Architecture Overview

The platform ingests highly fragmented biomedical, clinical, financial, demographic, and regulatory data, transforming it into a highly normalized Biomedical Knowledge Graph (BKG).

1.1. Core Tech Stack

- **Data Lake / Raw Storage:** Amazon S3 (JSON payloads, scraped HTML, legacy PDFs).
- **Relational/Geospatial Database:** PostgreSQL with PostGIS (for demographic/NFHS-5 heatmaps).
- **Graph Database:** Neo4j or Amazon Neptune (Property Graph for multi-hop semantic querying).
- **NLP / Computer Vision:** BERN2 (NER), DocTR (OCR), LayoutLM (VIE), Scispacy (Entity Linking).
- **Message Broker / Queueing:** Redis / Celery (Strictly required for API rate-limiting).

Data Layer	Primary Source	Role in SoW
Technical Merit	PubMed / Patents	Publication & Patent analysis for scouting.
Clinical Readiness	CTRI	Determining "Feasibility" (Phase of trial).
Market Viability	Tracxn / MCA	ROI, revenue potential, and competitor mapping.
Impact / Need	NFHS / NDAP	"Desirability" – mapping innovation to actual

		disease burden.
Regulatory Path	CDSCO / SUGAM	Regulatory analysis for novel therapies.

2. Data Ingestion & Transformation Pipelines

Layer 1: Technical Merit (PubMed & Global Patents)

- **Source:** NCBI PubMed, EPO OPS, USPTO Bulk Data, WIPO API.
- **Extraction Method:**
 - **PubMed:** NCBI E-utilities (REST/XML).¹ *Constraint:* Hard rate limit of 3 requests/second; implementation of asynchronous task queues is mandatory to avoid IP blocks.¹
 - **Patents:** EPO Open Patent Services (OPS) REST API for bibliographic data, legal status, and family structures.²
- **Transformation:** Implement the PubMed Knowledge Graph (PKG 2.0) methodology.³ Use BERN2 (multi-task NER) to extract bio-entities (genes, diseases, compounds) and link them between academic literature and patent claims.³

Layer 2: Clinical Readiness (CTRI & ClinicalTrials.gov)

- **Source:** Clinical Trials Registry - India (CTRI) & ClinicalTrials.gov (CTG).
- **Extraction Method:**
 - **CTRI:** No developer API exists. Requires Python/Selenium web scraping to dump HTML to local SQLite/PostgreSQL.⁴ Updates run every 4 weeks.⁵
- **Transformation:**
 - Regex parsing to extract secondary IDs (e.g., NCT numbers) to cross-reference CTRI records with the CTG database.⁶
 - Map free-text CTRI fields to standard CTG protocol definitions to resolve internal inconsistencies.

Layer 3: Market Viability (Tracxn & MCA V3)

- **Source:** Tracxn API, Ministry of Corporate Affairs (MCA) V3 Portal API.
- **Extraction Method:**
 - **Tracxn:** REST API (JSON) and scheduled SFTP data dumps (for offline analysis).⁷
 - **MCA V3:** Direct MCA Data API utilizing Corporate Identification Numbers (CIN).⁸ Extract modern XBRL filings for Forms AOC-4 (financials) and MGT-7 (annual returns).⁹
- **Transformation:** Standardize startup funding stages, capitalization tables, and revenue models using Tracxn's sector taxonomy.¹¹

Layer 4: Desirability & Impact (NDAP & NFHS-5)

- **Source:** National Data and Analytics Platform (NDAP).
- **Extraction Method:** NDAP open APIs.¹²
- **Transformation:** Merge Primary Population Census 2011 (denominators) with NFHS-5 Clinical, Anthropometric, and Biochemical (CAB) indicators.¹² Aggregate spatially down to the district level for disease burden heatmaps.

Layer 5: Regulatory Path (CDSCO SUGAM)

- **Source:** CDSCO SUGAM portal and upcoming Digital Drugs Regulatory System (DDRS).
- **Extraction Method:**
 - **Current State:** Scraping legacy, unstructured PDFs from SUGAM.
 - **Future State:** Transition to DDRS open APIs as CDSCO deprecates legacy portals.¹⁴
- **Transformation (PDF Processing Pipeline):**
 - Deploy **DocTR** for bounding-box OCR on visually rich documents.¹⁶
 - Pass coordinates to **LayoutLM** for layout-aware Visual Information Extraction (VIE) to extract key-value pairs (e.g., trial sites, adverse events).¹⁷
 - Use LLM/RAG orchestrations to interpret historical adverse drug events

(ADE) from parsed text.¹⁸

3. Biomedical Knowledge Graph (BKG) Schema

To execute multi-domain queries (e.g., finding under-funded startups targeting high-burden diseases), the data lake must be normalized into a graph schema mapping Subject-Predicate-Object triplets.

Entity Resolution & Normalization Keys

- **Physiological Entities:** Map extracted text spans (using Apache cTAKES/Scispacy) to the UMLS Metathesaurus. This links PubMed's MeSH terms directly to NDAP/NFHS-5's ICD-10 codings.¹⁹
- **Pharmacological Entities:** Utilize **DrugBank IDs** as the primary foreign key to join chemical structures in pharmaceutical patents to specific interventions tested in clinical trials.²⁰

Graph Schema Elements

Node Type	Source	Properties	Directed Edges
BioEntity	PubMed/PKG 2.0	Type (Gene, Disease), UMLS CUI	/
Trial	CTRI/CTG	Phase, Status, NCT_ID	/
Patent	USPTO/EPO	Filing_Date, Claims	/
Company	Tracxn/MCA	CIN, Stage, Funding	/
District	NDAP/NFHS-5	Geo_Coords, Prevalence_Rate	^^

4. Front-End Architecture & UI/UX Requirements

To enable users to effectively parse the massive scale of the Biomedical Knowledge Graph without experiencing cognitive overload, the platform implements an AI-driven, highly visual front-end architecture.

4.1. Natural Language Querying (RAG)

- The semantic query layer utilizes Retrieval-Augmented Generation (RAG) coupled with vector search.
- Users interact with the system via natural language prompts rather than complex graph query languages. The RAG pipeline integrates large language models (LLMs) to retrieve specific, context-aware insights across all integrated domains.
- This enables users to instantly access market research, competitive intelligence, and clinical data (e.g., mapping a clinical trial's eligibility criteria) via a simple chat interface.

4.2. Interactive Data Visualizations

- To communicate key insights clearly, the platform aggregates data into dynamic visual dashboards focusing on key performance indicators (KPIs).
- Visual components include **Scatter Plots** to uncover correlations between variables, **Heat Maps** to highlight patterns in site performance metrics or disease burden, and **Geospatial Maps** to visualize geographic trends in clinical trial enrollment or health outcomes across different districts.

4.3. Agentic AI Workflows

- The UI is powered by backend autonomous agents that act as intelligent workflow coordinators.
- These agents continuously monitor incoming data streams (such as new MCA filings or CTRI trial updates), extract structured information using natural language processing, validate it, and deliver proactive insights directly to operational dashboards.
- This architecture supports autonomous decision-making and coordinated task execution, drastically reducing manual data monitoring for the end-user.

4.4. Customized, Role-Based Dashboards

- The platform features highly customized dashboards allowing different personas to track particular health or financial metrics aligned with their specific goals.
- The primary interface utilizes a multi-functional "smart search" box that

provides advanced filtering options, enabling users to instantly retrieve patient demographics, startup data, or clinical trial statuses based on recent activities and personalized criteria.

5. Compliance & Data Governance Requirements

The platform architecture is strictly bound by India's Digital Personal Data Protection (DPDP) Act of 2023.²¹

- **PII & Data Handling:** Clinical trial investigator names, CDSCO regulatory contacts, and startup founder profiles constitute digital personal data. The platform acts as a Data Fiduciary.²²
- **Data Retention Rules:** Must comply with DPDP one-year minimum retention rules for system logs and traffic data required for lawful audits, after which automated anonymization/deletion scripts must execute unless another law mandates longer retention.²³
- **Data Residency:** While cross-border transfer is permitted to non-restricted countries, the Indian Government holds notification rights to block specific transfers. Core infrastructure (PostgreSQL, Neo4j, S3) must be hosted on an India-based Cloud Region (e.g., AWS ap-south-1) to future-proof against sovereign localization mandates.²⁴
- **Security Architecture:** End-to-end payload encryption (TLS 1.3), AES-256 for data at rest, and strict Role-Based Access Controls (RBAC) to ensure multi-tenant data isolation.²² Fines for non-compliance can reach ₹250 crore.²⁶

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